Problem 1. Let $P(x, y)$ be the statement "Student x has taken class y, "where the domain for x consists of all students in your class and y consists of all computer science courses at your school. Express each of these quantifications in english.

a) $\exists x \exists y\, P(x, y)$
b) $\exists x \forall y\, P(x, y)$
c) $\forall x \exists y\, P(x, y)$
d) $\exists y \forall x\, P(x, y)$
e) $\forall y \exists x\, P(x, y)$
f) $\forall x \forall y\, P(x, y)$

Answer.

a) There is at least one student in my class that took at least one of the computer science courses at my school.
b) There is at least one student in my class that has taken all of the computer science courses at my school.
c) All the students in my class has taken at least one of the computer science courses at my school.
d) There is at least one computer science course at my school that has been taken by all the students in my class.
e) All of the computer science courses has been taken by at least one student in my class.
f) All of the students in my class have taken all of the computer science courses at my school.

Problem 2. Let $Q(x, y)$ be the statement “Student x has been a contestant on quiz show y.” Express such of these sentences in terms of $Q(x, y)$, quantifiers, and logical connectives, where the domain for x consists of all students at your school and for y consists of all quiz shows on television.

a) There is a student at your school who has been a contestant on a television quiz show.
b) No student at your school has ever been a contestant on a television quiz show.
c) There is a student at your school who has been a contestant on *Jeopardy!* and on *Wheel of Fortune*.
d) Every television quiz show has had a student from your school as a contestant.
e) At least two students from your school have been contestants on *Jeopardy!*.

Answer.

a) $\exists x \exists y\, Q(x, y)$
b) $\neg(\exists x \exists y\, Q(x, y))$
c) $\exists x\, (Q(x, \text{"Jeopardy!"}) \land Q(x, \text{"Wheel of Fortune"}))$
d) $\forall y \exists x\, Q(x, y)$
e) $\exists x_1 \exists x_2\, ((x_1 \not\equiv x_2) \land Q(x_1, \text{"Jeopardy!"}) \land Q(x_2, \text{"Jeopardy!"}))$

Problem 3. Express each of these statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain consists of all integers.

a) The product of two negative integers is positive.
b) The average of two positive integers is positive.
c) The difference of two negative integers is not necessarily negative.
d) The absolute value of the sum of two integers does not exceed the sum of the absolute values of these integers.

Answer.

a) $\forall x \forall y\,(((x<0)\land(y<0))\to(xy>0))$
b) $\forall x \forall y\,\big(((x>0)\land(y>0))\to\big(\frac{x+y}{2}>0\big)\big)$
c) $\exists x \exists y\,(((x<0)\land(y<0))\land(x-y\geq 0))$
d) $\forall x \forall y\,(|x+y|\leq|x|+|y|)$

Problem 4. Find a common domain for the variables x, y, and z for which the statement $\forall x \forall y\,((x \neq y) \to \forall z\,((z = x) \vee (z = y)))$ is true and another domain for which it is false.

Answer.

- **True**: A set that contains just one or item. Since $(x \neq y)$ (the left side of the implication) is always false, this statement is always true.
- **False**: Natural numbers. Two different numbers x and y does not imply that all rest of the numbers is either x or y.

Problem 5. What rule of inference is used in each of these arguments?

a) Kangaroos live in Australia and are marsupials.
Therefore, kangaroos are marsupials.
b) It is either hotter than 100 degrees today or the pollution is dangerous.
It is less than 100 degrees outside today.
Therefore, the pollution is dangerous.
c) Linda is an excellent swimmer.
If Linda is an excellent swimmer, then she can work as a lifeguard.
Therefore, Linda can work as a lifeguard.
d) Steve will work at a computer company this summer.
Therefore, this summer Steve will work at a computer company or he will be a beach bum.
e) If I work all night on this homework, then I can answer all the exercises.
If I answer all the exercises, I will understand the material.
Therefore, if I work all night on this homework, then I will understand the material.

Answer.
a) Simplification, $(p \land q) \to q$
b) Disjunctive syllogism, $((p \lor q) \land \neg p) \to q$
c) Modus ponens, $(p \land (p \to q)) \to q$
d) Addition, $p \to (p \lor q)$
e) Hypothetical syllogism, $((p \to q) \land (q \to r)) \to (p \to r)$

Problem 6. For each of these arguments, explain which rules of inference are used for each step.

a) "Linda, a student in this class, owns a red convertible. Everyone who owns a red convertible has gotten at least one speeding ticket. Therefore, someone in this class has gotten a speeding ticket."
b) "Each of five roommates, Melissa, Aaron, Ralph, Veneesha, and Keeshawn, has taken a course in discrete mathematics. Every student who has taken a course in discrete mathematics can take a course in algorithms. Therefore, all five roommates can take a course in algorithms next year."
c) "All movies produced by John Sayles are wonderful. John Sayles produced a movie about coal miners. Therefore, there is a wonderful movie about coal miners."
d) "There is someone in this class who has been to France. Everyone who goes to France visits the Louvre. Therefore, someone in this class has visited the Louvre."

Answer.

a) Linda is in this class and she owns a red convertible. Therefore, she must have gotten at least one speeding ticket, by universal instantiation from the second statement. Since Linda is in this class, we use existential instantiation to say that at least one person in the class has gotten a speeding ticket.
b) Since all five roommates (our domain) has taken discrete mathematics, we use universal generalization to say that all of them can pass the predicate that $P(x) \equiv$ "x has taken a course in discrete mathematics". Since any student that has taken discrete mathematics can take an algorithm course, all of our roommates can take it by modus ponens.
c) We first use modus ponens to say that the movie John sayles produced about coal miners is wonderful. Then, we use existential generalization to say that there is a wonderful movie about coal miners.
d) We use hypothetical syllogism and existential generalization.

Problem 7. For each of these arguments determine whether the argument is correct or incorrect and explain why.

a) Everyone enrolled in the university has lived in a dormitory. Mia has never lived in a dormitory. Therefore, Mia is not enrolled in the university.
b) A convertible car is fun to drive. Isaac's car is not a convertible. Therefore, Isaac's car is not fun to drive.
c) Quincy likes all action movies. Quincy likes the movie *Eight Men Out*. Therefore, *Eight Men Out* is an action movie.
d) All lobstermen set at least a dozen traps. Hamilton is a lobsterman. Therefore Hamilton sets at least a dozen traps.

Answer.

a) It is correct. It is a universal statement (everyone), so if someone who haven't lived in a dormitory, they are not in the university.
b) Incorrect. There can exist a car where it is not a convertible and it is fun to drive. The first statement is only existential.
c) Incorrect. Quincy can like movies that is not the action genre.
d) Correct. We can deduct that by modus ponens.

Problem 8. What is wrong with this statement? Let $S(x, y)$ be "x is shorter than y." Given the premise $\exists s\, S(s, \text{Max})$, it follows that $S(\text{Max}, \text{Max})$. Then by existential generalization it follows that it follows that $\exists x\, S(x, x)$, so that someone is shorter than himself.

Answer. We can't say that "Max is shorter than Max", since we can't select any arbitrary value for an existential instantiation. So the statement that $S(\text{Max}, \text{Max})$ is wrong, and the following existential generalization is also wrong.

Problem 9. Use rules of inference to show that if $\forall x\,(P(x) \lor Q(x))$ and $\forall x\,((\neg P(x) \land Q(x)) \to R(x))$ are true, then $\forall x(\neg R(x) \to P(x))$ is also true, where the domains of all quantifiers are the same.

Answer.

We have:

$$\begin{aligned} P(c) \lor Q(c) &\qquad \text{universal instantiation} \\ (\neg P(c) \land Q(c)) \to R(c) &\qquad \text{universal instantiation} \end{aligned}$$

We assume $\neg R(c)$:

$$\begin{aligned} (\neg P(c) \land Q(c)) \to R(c) & \\ \neg R(c) & \\ \neg(\neg P(c) \land Q(c)) &\qquad \text{modus tollens} \\ P(c) \lor \neg Q(c) &\qquad \text{demorgan, double negation} \end{aligned}$$

Bringing it back:

$$\begin{aligned} P(c) \lor \neg Q(c) & \\ P(c) \lor Q(c) & \\ P(c) &\qquad \text{resolution} \end{aligned}$$

We now have proved $P(c)$ by assuming $\neg R(c)$. By universal generalization, since we have $P(c)$ for an arbitrary c, we can say that $\forall x\,(\neg R(x) \to P(x))$ is true. ■

Challenge 1. Show that $\forall x\, P(x) \lor \forall x\, Q(x)$ and $\forall x \forall y\, (P(x) \lor Q(y))$, where all quantifiers have the same nonempty domain, are logically equivalent. (The new variable y is used to combine the quantifications correctly.)

Answer.

We need to prove that $(\forall x\, P(x) \lor \forall x\, Q(x)) \Leftrightarrow (\forall x \forall y\, (P(x) \lor Q(y)))$.

Direction 1: $(\forall x P(x) \lor \forall x Q(x)) \rightarrow (\forall x \forall y (P(x) \lor Q(y)))$

We assume that $\forall x P(x) \lor \forall x Q(x)$, which gives us two cases:

Case 1:

$$\begin{array}{rl} P(c) & \text{universal instantiation} \\ P(c) \lor Q(d) & \text{addition} \end{array}$$

Case 2:

$$\begin{array}{rl} Q(c) & \text{universal instantiation} \\ Q(c) \lor P(d) & \text{addition} \end{array}$$

Combining the results of the addition:

$$\begin{array}{ll} P(c) \lor Q(d) & \\ Q(c) \lor P(d) & \\ P(c) \lor Q(d) & \text{simplification} \end{array}$$

We now have $P(c) \lor Q(d)$, and by universal generalization, we can say that we have $\forall x \forall y\, (P(x) \lor Q(y))$.

Direction 2: $(\forall x \forall y\, (P(x) \lor Q(y))) \rightarrow (\forall x\, P(x) \lor \forall x\, Q(x))$

$$\begin{array}{rl} \neg(\forall x P(x) \lor \forall x Q(x)) \rightarrow \neg(\forall x \forall y (P(x) \lor Q(y))) & \text{contraposition} \\ \exists x \neg P(x) \land \exists x \neg Q(x) \rightarrow \exists x \exists y \neg(P(x) \lor Q(y)) & \text{negate, demorgan} \\ \exists x \neg P(x) \land \exists x \neg Q(x) \rightarrow \exists x \exists y \neg P(x) \land \neg Q(y) & \text{demorgan again} \\ \neg P(c) \land \neg Q(d) \rightarrow \neg P(c) \land Q(d) & \text{existential instantiation both sides} \end{array}$$

■